WORKED SOLUTION

STEP I 2005, Question 6

Pure • Coordinate geometry

STEP I 2005, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q6

The key idea

A condition like $AP = 2BP$ involves square roots, so square it first: distances squared are polynomials, and the $x^2 + y^2$ terms come with the same coefficient on both sides, which is why the path is a circle. In part (ii) we do the same for a general pair of points on the x-axis, and then two equations of the form $x^2 + y^2 + px + q = 0$ describe the same circle exactly when p and q agree. Comparing coefficients gives two expressions for k^2, and eliminating k gives both results.

Solution

Part (i)

Both lengths are non-negative, so $AP = 2BP$ is equivalent to $AP^2 = 4BP^2$:

$$(x-5)^2 + (y-16)^2 = 4[(x+4)^2 + (y-4)^2].$$

Expanding,

$$x^2 - 10x + y^2 - 32y + 281 = 4x^2 + 32x + 4y^2 - 32y + 128.$$

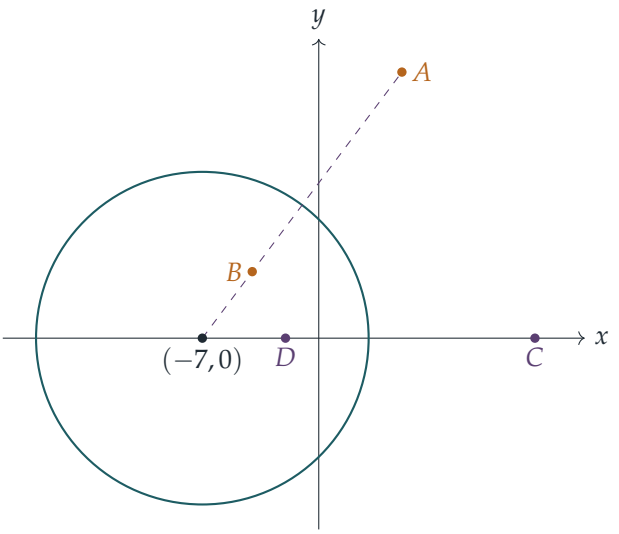
The y terms cancel, and collecting everything on the right gives

$$3x^2 + 42x + 3y^2 - 153 = 0, \qquad \text{that is,} \qquad x^2 + 14x + y^2 - 51 = 0.$$

Completing the square, $x^2 + 14x = (x+7)^2 - 49$, so

$$(x+7)^2 + y^2 = 100,$$

as required. The path is the circle with centre $(-7, 0)$ and radius 10.



(The points $C = (13, 0)$ and $D = (-2, 0)$ are one possible pair for part (ii), with $k = 2$. Notice that A, B and the centre are collinear too.)

Part (ii)

As before, $QC = kQD$ is equivalent to $QC^2 = k^2QD^2$. With $Q = (x, y)$,

$$(x-a)^2 + y^2 = k^2[(x-b)^2 + y^2].$$

Collecting terms on the right,

$$(k^2-1)(x^2+y^2) - 2(k^2b - a)x + (k^2b^2 - a^2) = 0.$$

Since $k > 1$, we have $k^2 - 1 \neq 0$ and we may divide by it:

$$x^2 + y^2 - \frac{2(k^2b-a)}{k^2-1}\,x + \frac{k^2b^2 - a^2}{k^2-1} = 0.$$

Ask yourself Why are we allowed to compare coefficients with $x^2 + 14x + y^2 - 51 = 0$?

An equation $x^2 + y^2 + px + q = 0$ can be rewritten as $(x + \frac{1}{2}p)^2 + y^2 = \frac{1}{4}p^2 - q$. If it is the circle with centre $(-7, 0)$ and radius 10, then the centre forces $p = 14$ and the radius forces $\frac{1}{4}p^2 - q = 100$, so $q = -51$. A circle has only one centre and one radius, so the two equations must have the same p and the same q.

Comparing the coefficients of x:

$$\frac{k^2b - a}{k^2 - 1} = -7 \implies k^2b - a = -7k^2 + 7 \implies k^2(b+7) = a + 7.$$

Comparing the constant terms:

$$\frac{k^2b^2-a^2}{k^2-1}=-51 \implies k^2b^2-a^2=-51k^2+51 \implies k^2(b^2+51)=a^2+51.$$

Now $b^2+51>0$, and $b+7\neq 0$: if $b=-7$, the first equation would give $a=-7=b$, which is ruled out. So we can divide by both to get two expressions for k^2:

$$\frac{a+7}{b+7}=k^2=\frac{a^2+51}{b^2+51},$$

which is the first result.

For the second, cross-multiply:

$$(a+7)(b^2+51)=(b+7)(a^2+51).$$

Expanding, the constant 357 cancels from both sides and we are left with

$$ab^2+51a+7b^2=a^2b+51b+7a^2.$$

Group the terms so that $a-b$ appears in each:

$$\begin{aligned}0&=(a^2b-ab^2)-51(a-b)+7(a^2-b^2)\\&=(a-b)[ab-51+7(a+b)].\end{aligned}$$

Since $a\neq b$, we can divide by $a-b$, so

$$ab+7a+7b=51, \quad \text{so} \quad ab+7a+7b+49=100, \quad \text{that is,} \quad (a+7)(b+7)=100,$$

as required.

Checks

- Try $b=-2$: then $b+7=5$, so $a+7=20$ and $a=13$, giving $k^2=20/5=4$. The point $(3,0)$ is on the circle, and $QC=10=2\times 5=2\,QD$. The point $(-17,0)$ is on it too, with $QC=30=2\times 15$.
- In part (i), the point $(3,0)$ gives $AP=\sqrt{4+256}=\sqrt{260}$ and $BP=\sqrt{49+16}=\sqrt{65}$, and indeed $\sqrt{260}=2\sqrt{65}$.
- We also checked numerically that random points on the circle satisfy $QC=kQD$ for many pairs with $(a+7)(b+7)=100$, and that every pair with $a\neq b$ satisfying the first result also satisfies the second.

Common slips

- Comparing coefficients before dividing by k^2-1: the two equations only need to agree up to an overall factor.
- Dividing by $b+7$ without checking it is non-zero.
- Losing the factor $a-b$: it is exactly the trivial solution $a=b$ that the question excludes.

Beyond STEP

The locus of points whose distances from two fixed points are in a fixed ratio $k \neq 1$ is a *circle of Apollonius*. The result $(a+7)(b+7) = 100$ says that C and D are on the same side of the centre with $MC \cdot MD = r^2$, where M is the centre and r the radius: C and D are *inverse points* in the circle. (So are A and B: $MA = 20$, $MB = 5$.) Inversion in a circle is a powerful tool in geometry and in complex analysis, where it is the map $z \mapsto 1/\bar{z}$; Needham's *Visual Complex Analysis* has a beautiful account.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.